

Semruk: A Scale-Invariant, Consistency-First Agent for the ANAC 2026 Human-Agent Negotiation League

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Abstract

We present *Semruk*, our submission to the Human-Agent Negotiation (HAN) league of ANAC 2026. *Semruk* is a NegMAS BOA agent — a Boulware concession curve with a Pareto-aware bid-selection rule, a Smith-style frequency opponent model that exposes the `opponent_ufun` the deception score reads, and a phase-conditioned text module that adds no LLM call on the critical path. The problem we set out to solve was **cross-scenario rank instability**: with absolute thresholds, the same parameter behaved differently on a $[0, 1]$ scenario than on a $[0, 100]$ one, and our rank swung between near-top and last as scenarios were sampled. We fix it with (i) **scale-invariant thresholds** (every threshold is a fraction of attainable surplus, so one number behaves identically everywhere); (ii) a **rational final-step acceptance net** that removes the deadlock-zero tail and makes a tough 0.92 floor safe; (iii) a **non-degenerate cold-start opponent-model prior** that recovers a full deception point against eager acceptors at zero advantage cost; and (iv) **human-perception-tuned messages**. Versions were chosen by variance-aware round-robin tournaments judged on worst-case and cross-scenario standard deviation. Our first server submission placed 3rd; the final agent is the most consistent across a nine-version, eleven-scenario round-robin. We also report four negative results that bound where further tuning cannot help.

1 Introduction

The HAN league asks an agent to negotiate bilaterally with a *human* partner under the alternating-offers protocol (AOP) augmented with free-form text. The competition runs in two stages: an LLM round against personality-adjusted verified LLM negotiators that simulate humans, and a final human round with crowd-sourced partners. Agents are ranked on the utility they achieve against human subjects and on the perception of human partners measured by questionnaires [5].

Two features of this setting shaped our work. First, the evaluation field is heterogeneous and partly unknown: the harness samples scenarios of different sizes, utility ranges and reserved values, and partners range from stubborn to eager. An agent that is excellent on one scenario family but poor on another sees its *rank* swing from run to run. Our overriding goal was therefore not peak score but **consistency** — be good in *every* scenario, with a high worst case and low variance. Second, evaluation is human-facing and multi-objective: an agent that wins more utility but reads as cold or manipulative can still

lose on the questionnaire, so the natural-language layer is a scored channel.

2 Background and Scoring

All ANAC 2026 leagues are built on NegMAS [4], whose modular BOA architecture decomposes a negotiator into a **Bidding strategy**, an **Opponent model** and an **Acceptance strategy** [2]. We adopt this decomposition because each component is testable in isolation and because the deception term is computed from the `opponent_ufun` the wrapper exposes.

The automated metric gives negotiator i the score

$$\text{score}_i = \underbrace{\frac{u_i(\omega^*) - r_i}{u_i^{\max} - r_i}}_{\text{advantage} \in [0,1]} + \underbrace{\frac{\alpha_i}{\sum_j \alpha_j}}_{\text{deception} \in [0,1]}, \quad (1)$$

where ω^* is the agreement, r_i the reserved value and $\alpha_i = \frac{1}{2}(1 + \tau_K(u_{\text{opp}}, \hat{u}_{\text{opp}}^{(i)}))$ is the Kendall- τ optimality of i 's own model of its opponent. A point easily missed is that advantage is **normalised per scenario**: a $[0, 100]$ scenario does not dominate a $[0, 1]$ one, so both halves of the score live in $[0, 1]$ and matter equally. The deception term is a normalised split of one modelling point; raising it is equivalent to modelling the opponent better or being harder to model.

3 Agent Design

Semruk has two layers. The inner core (`_SemrukCore`) extends NegMAS's `BOANegotiator` with a custom offering policy, a layered acceptance rule and a frequency opponent model. The outer adapter (`SemrukNegotiator`, the registered class) extends `SAOMetaNegotiator`: it attaches a phase-conditioned message to every move and updates the opponent model, because for adapted negotiators NegMAS does not reliably fire the BOA-internal callback.

Scale-invariant thresholds. Every threshold is expressed against the normalised utility

$$\nu(\omega) = \frac{u(\omega) - r}{u^{\max} - r} \in [0, 1], \quad (2)$$

the fraction of attainable surplus above walking away. A floor of 0.92 thus means “92% of my reachable surplus” in *every* scenario, regardless of scale or reserved value.

Algorithm 1 Scale-invariant Pareto-aware bid selection.

```
1:  $a \leftarrow \max((1-t)^4, f(t))$  // Eq. 3, in  $\nu$ 
2:  $C \leftarrow \{\omega : a \leq \nu(\omega) \leq \min(1, a+0.10+0.30t)\}$ 
3: if  $|C| < 5$  then  $C \leftarrow$  top outcomes of  $\Omega$  by  $u$ 
4: sort  $C$  desc. by  $(\hat{u}_{\text{opp}}, u)$ 
5: if  $\text{rand}() < p_{\text{decep}}(t)$  then  $\omega^* \leftarrow$  sample top-8
6: else  $\omega^* \leftarrow C[0]$ 
7: if  $\omega^*$  repeats last bid then take next distinct bid
8: return  $\omega^*$ 
```

Bidding. The policy clamps a Boulware ($e=4$) curve [3] from below by a deadline-aware floor that starts at $\underline{a}=0.92$ (in ν -units) and decays in three phases, holding firm early and conceding only late:

$$f(t) = \begin{cases} \underline{a} & t < p_1 \\ \underline{a}(1 - 0.6 \frac{t-p_1}{p_2-p_1}) & p_1 \leq t < p_2 \\ \underline{a}(0.4 - 0.3 \frac{t-p_2}{1-p_2}) & t \geq p_2, \end{cases} \quad (3)$$

with $p_1=0.93$, $p_2=0.97$. Within a growing window $\nu \in [a(t), a(t) + 0.10 + 0.30t]$, candidates are ranked lexicographically by (estimated opponent utility, own utility); we propose the top bid, or with probability $p_{\text{decep}}(t)$ sample from the top eight (Alg. 1).

Acceptance: a rational net. We start from **AC_{Next}** [1] — accept iff the offer is at least as good as our next bid — and add an endgame clause and, crucially, a **rational final-step clause** that accepts *any* strictly-positive-advantage offer once $t \geq 0.985$. This makes a tough 0.92 floor safe: against a stubborn partner we never walk away with zero to defend the floor; we take a small positive deal. The deadlock-zero tail it removes was our largest source of variance.

Opponent model. A Smith-style frequency model [6], kept as a real `UtilityFunction`, estimates per-issue value scores from proposal counts and issue weights from their concentration, so the scorer can compute Kendall’s τ for the deception term.

4 Original Contributions

(N1) Scale-invariance + rational acceptance. Earlier versions used *absolute* thresholds; on a $[0, 100]$ scenario a floor of 0.55 is no floor at all, while on $[0, 1]$ it is a hard one, so the same parameter produced wildly different behaviour and our rank swung run to run. Expressing every threshold against ν makes one number behave identically everywhere. Rather than lower the floor to avoid deadlocks, we *raised* it to 0.92 and made it safe with the rational net — a tough, scale-invariant offerer paired with a reasonable, deadlock-proof acceptor. This pair is the design most responsible for our consistency.

(N2) A non-degenerate cold-start prior. When a partner accepts our opening offer before ever counter-offering, the

Table 1: Cross-scenario robustness over 11 scenarios (normalised; higher Worst/Robust, lower Std = steadier rank).

Version	Mean	Worst	Std	Robust
v0.9	6727	3831	1980	4747
v0.10	6676	3722	2016	4660
v0.8	6375	3189	1955	4420
v0.7	6266	2392	2097	4169

frequency model has seen nothing, returns a constant, and Kendall’s τ over a constant vector is undefined ($\rightarrow -1$), zeroing our normalised deception share even though that partner did no modelling of us — a full point forfeited in exactly the games we already win on advantage. The fix is a **cooperative cold-start prior**: until a real opponent offer arrives, the model ranks outcomes by *our own* `ufun`. This keeps the model non-degenerate (deception ≈ 1.0 vs eager partners) while leaving the bid byte-for-byte unchanged, since the offering policy already breaks opponent-utility ties by our own utility. The prior is overwritten on the first observed offer, so modelling partners are unaffected — a deception gain at *zero* advantage cost.

(N3) Human-perception-tuned messaging. Because the human-round ranking includes a perception questionnaire, the message is a scored channel. We rewrote the templates to be brief, warm, fairness-framed, reason-giving, and collaborative rather than threatening near the deadline. The messages are generated *after* the base negotiator chooses its move, so they cannot change the automated score: a pure addition for the human stage.

5 Experiments

Methodology. Our harness loads the three official scenarios plus a generator of diverse synthetic ones (varied opposition, reserved value, scale), normalises each side’s advantage onto a $[0, 10\,000]$ scale, and runs a full round-robin among all tagged versions and the non-LLM baselines. Because our problem is rank *instability*, we select on a variance-aware criterion — high worst-case and low cross-scenario standard deviation, summarised as **Robust = Mean – Std** — and re-run under a second seed to guard against overfitting.

Consistency. Our first version placed 3rd in a public server tournament. Table 1 gives the robustness view of a nine-version, eleven-scenario round-robin (3564 negotiations): the worst-case score climbs from 2392 (v0.7) to 3831 (v0.9) as the standard deviation falls, so the scale-invariant, rational-acceptance versions no longer finish last on a bad draw. v0.10 ties v0.9 on advantage (the cold-start prior is advantage-neutral) and v0.11 inherits that profile exactly (text-only); a second seed reproduced the same top tier.

Table 2: Deception share before/after the cold-start prior. Advantage is unchanged in every row.

Partner	before	after
Nice (eager acceptor)	0.00	1.00
Boulware (non-modeller)	1.00	1.00
BOANeg / self (modellers)	0.51	0.51

Deception. Table 2 isolates the cold-start fix (N2): against an eager “Nice” partner the deception share jumps from 0.00 to 1.00 with advantage unchanged; the tough non-modeller and the modelling partners are unaffected. Against equal modellers the share sits near 0.51; a short derivation from Eq. (1) shows even a perfect model raises it only to ≈ 0.54 , so chasing it by sacrificing advantage is a poor trade.

Negative results. Four directions showed *no* headroom and were deliberately *not* added, sparing the agent needless variance: **(a) opponent-type floor adaptation** — an oracle gains ≤ 61 points while one mis-classification costs ≈ 629 , so any real classifier has negative expected value; **(b) offer-diversification deception** — sweeping the exploration probability $0 \rightarrow 0.9$ leaves modelability and deception flat, because a high floor forces our preferred values on high-salience issues; **(c) per-scenario floor adaptation** — our weakest *relative* scenario (Island) actually prefers a *higher* floor; and **(d) acceptance/concession micro-tuning** — a three-seed sweep beats the baseline’s Robust by $\leq 1.5\%$, within noise.

6 Discussion and Limitations

The measurable, agent-vs-agent half of the problem is at a well-validated optimum, and the negative results are part of the contribution: they explain why the final agent deliberately omits an opponent classifier and an offer-diversification scheme that earlier versions carried, both of which add variance without adding score. The one aspect we could not close experimentally is the **human-perception channel**: text impact cannot be measured on our hardware (the automated metric ignores text, the template LLM decides on offers only, and we have no human in the loop), so our v0.11 messaging is a principled, advantage-neutral choice rather than a measured gain. Evaluating against a text-reading human-simulating LLM, or against humans in the HANI GUI, is the direction that would most benefit from further work.

7 Conclusion

For a competition that samples scenarios at random, cross-scenario *stability* — not peak score — is what protects rank, and that stability comes from making thresholds scale-invariant and from never letting a tough stance turn into a zero. The combination of scale-invariant Boulware bidding, a rational acceptance net, a non-degenerate cold-start opponent

model and a perception-tuned text layer is, to our knowledge, novel for the HAN league, and yields the most consistent agent we built.

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